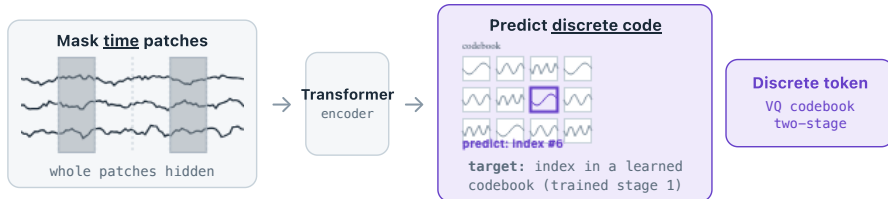


LaBraM

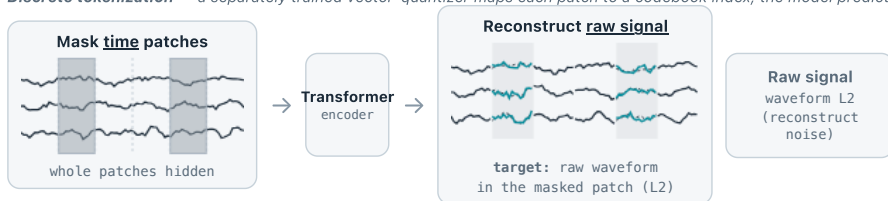
VQ codebook
(2-stage)



Discrete tokenization — a separately trained vector-quantizer maps each patch to a codebook index; the model predicts that index.

CBraMod

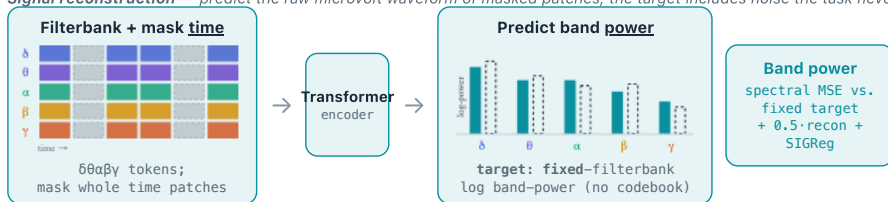
masked
reconstruction



Signal reconstruction — predict the raw microvolt waveform of masked patches; the target includes noise the task never uses.

Ours

codebook-free
spectral



Spectral prediction (ours) — a learnable filterbank tokenizes each patch into bands; predict the masked patches' log band-power against a **fixed** (non-learnable) filterbank target. Single-stage, no codebook; raw-signal reconstruction is a light auxiliary.

■ δ ■ θ ■ α ■ β ■ γ ■ masked patch ■ prediction ■ target